Transistor-Level Design and Validation of a Silicon-Carbide JFET Intel-4004-Compatible 4-bit Processor

Christian Garry* and Alton Horsfall†

***Abstract*—Silicon Carbide (SiC) Junction Field Effect Transistors (JFETs) are a leading candidate for digital logic operating in radiation environments and at temperatures beyond the reach of bulk silicon. To the authors' knowledge, this work presents the first transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor architecture, comprising 5537 transistors across the ALU, instruction decoder, program counter, three-level stack, and sixteen-register scratch pad, with a compatible 4001-style ROM model used for program execution. The contributions are three. First, a complete 4004-compatible transistor-level SiC JFET RTL design captured at the schematic level in LTspice. Second, a program-level co-validation methodology in which a custom Python reference emulator matches the LTspice CPU register-for-register at every instruction boundary, demonstrated on a bitwise AND subroutine from the MCS-4 programming manual and a 16-bit floating-point multiplication. Third, measured NAND/NOR primitive switching behaviour consistent in timing and waveform shape with LTspice simulation, characterised from 1 Hz up to 800 kHz (NOR) and 1 MHz (NAND), with an unresolved 1.5–2 V baseline offset and other limitations of the present validation chain treated explicitly.**

***Index Terms*—Silicon Carbide (SiC), Junction Field Effect Transistors (JFETs), Radiation-Environment Electronics, High-Temperature Electronics, Intel 4004 Architecture, Resistor–Transistor Logic, Circuit Simulation, Emulator Co-validation**

I. Introduction

COMPUTING in radiation- and temperature-extreme environments — fusion reactors, geothermal boreholes, deep-space probes — is constrained by the failure of silicon-based logic above 200–400 °C and under heavy ionising radiation. Wide-bandgap semiconductors offer a path forward, with Silicon Carbide (SiC) the most mature of the candidate materials: 4H-SiC has a bandgap of 3.26 eV [1], compared to 1.12 eV for silicon [2], and a high atomic-displacement threshold energy (24 eV for carbon and 66 eV for silicon in 4H-SiC [3], against 21 eV in bulk Si [4]). SiC Schottky-barrier diodes have been measured under proton [5], electron [6] and light-ion [7] irradiation without deviation from linear charge-collection behaviour, making the material suitable for both fusion and off-world applications.

Among SiC active devices, the junction field-effect transistor (JFET) is uniquely suited to high-temperature digital logic: it relies on a buried p-n junction rather than a metal-semiconductor interface, avoiding the Schottky-barrier degradation that disqualifies MESFETs above ∼300 °C [8]. SiC JFETs have been demonstrated as discrete devices at ∼1000 °C, with the main remaining degradation mechanism being interdiffusion of the metal ohmic contacts rather than wafer-level failure [9]. Complementary JFET (CJFET) logic gates have been built and measured to 623 K [10] and simulated to 500 °C [11]; the analysis in this paper instead uses resistor-transistor logic (RTL), since CJFET requires complementary p- and n-type devices and the available SiC JFET technology is single-polarity. RTL maps naturally to resistor-loaded pull-up/pull-down structures on single-polarity SiC JFETs. The comparison with the Intel 4004 is architectural rather than device-technological: the original 4004 was implemented in silicon-gate PMOS, whereas the present work implements a 4004-compatible architecture using SiC JFET RTL.

The Intel 4004 was chosen as the target architecture because of its small but complete ISA — 46 instructions, a 12-bit program counter, a 3-level subroutine stack, sixteen 4-bit scratch-pad registers, and an ALU restricted to 4-bit addition and accumulator rotation [12], [13]. This is large enough to be a realistic test of an integrated SiC JFET design (the present implementation totals 5537 JFETs) yet small enough that every subsystem can be hand-designed, validated against a Python reference emulator, and reasoned about in a single paper.

Table I situates this work against the existing SiC JFET digital-logic literature. To the authors' knowledge, this is the first transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor architecture; prior demonstrations have been at the logic-gate or small-IC scale. The contribution of this paper is therefore not a new device technology but a complete architecture-scale design captured in LTspice, paired with a validation methodology — register-level co-simulation against a Python reference emulator and measured-versus-simulated agreement of the primitive gates — that gives independent evidence that the simulated processor is functionally correct.

The paper is organised as follows. Section II describes the SiC JFET RTL logic family and the decomposition of the 4004 architecture into subsystems. Section III describes the design and validation methodology, including the LTspice transient simulation, the Python reference emulator, and the assembler/PWL toolchain. Section IV presents the program-level register-state co-validation and the measured-versus-simulated primitive-gate results. Section V delineates the limitations of

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TABLE I: SiC JFET digital-logic literature against which this work is positioned. "Validation" distinguishes simulation-only demonstrations from those backed by measured silicon. Operating temperature is the highest at which the cited work reports successful logic-gate operation; for this paper it is room temperature, since the device model used for the LTspice CPU simulations is not temperature-scaled.

Reference	Process / device	Logic family	Complexity	Max validated T	Validation
Zolper [8] (1998)	GaAs / SiC / GaN JFETs	–	review only	–	review
Raynaud [14] (1997)	6H-SiC buried-gate JFET	–	single device	300 °C	measured (device)
Neudeck [15] (2000)	6H-SiC JFET	RTL	individual gates	600 °C	measured
Neudeck [9] (2018)	4H-SiC JFET	RTL	small digital ICs	∼1000 °C	measured
Habib [11] (2013)	4H-SiC CJFET	CJFET	individual gates	500 °C	simulation
Kaneko [10] (2022)	4H-SiC CJFET	CJFET	logic gate	350 °C (623 K)	measured
This work	4H-SiC JFET (model)	RTL	4-bit CPU (5537 JFETs)	27 °C (model)	LTspice + emulator + measured primitives

the work, including the room-temperature device model and the measured voltage offset, and Section VI concludes.

II. Theory

In this paper, "4004-compatible" means ISA-level and program-level compatibility with the documented MCS-4 instruction semantics under the modelled 4001-style ROM interface. It does not imply pin-level, voltage-level, cycle-timing, fabrication-process, or electrical-bus equivalence with the original Intel 4004. In particular, the data-bus implementation described in Section II-B is adapted to the SiC JFET RTL logic family rather than copied electrically from the original PMOS device.

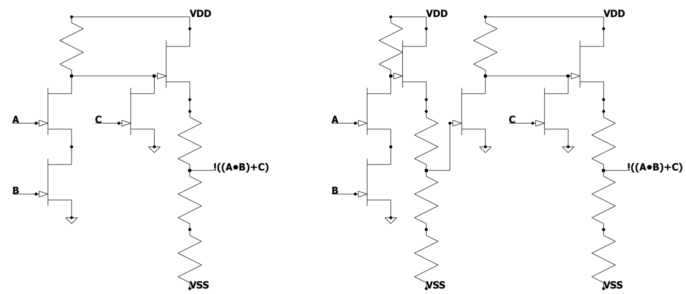

Fig. 2: Circuit diagram of a condensed boolean gate (Left) and a combination of base gates (Right), illustrating the transistor reduction for the boolean equation (A AND B) NOR C

A. Logic Primitives

The LTspice model of the JFET was used to create a logic primitive. This was extended to form more complex Logic Gates. Applying a voltage to the Gate of a JFET causes the depletion region to increase. This narrows the conducting channel increasing the resistance, which decreases the current that can pass through it. If the source of the JFET is connected to a voltage source through a resistor there will then be a change in voltage at the JFET's source when a voltage is applied to the Gate. This does not provide a large enough voltage swing between high and low and has implications for the fan out of the gate. Therefore, a voltage follower was used which works as a voltage level shifter and shifts the output voltage back to input levels of the logic gate (High: -0.8 V, Low: -3.6 V). These values were selected as they kept the output voltage from decaying or oscillating when used as inputs for subsequent gates whilst using standard resistor values.

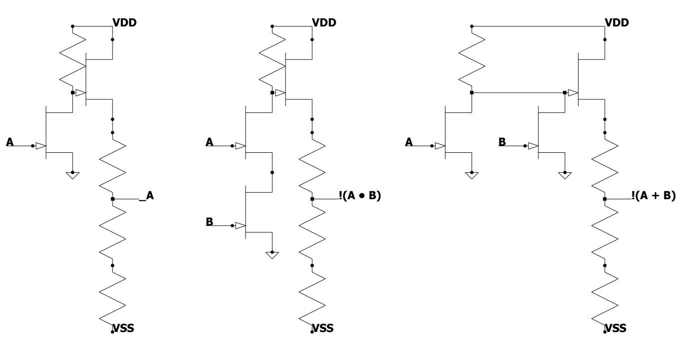

Fig. 1: Circuit diagrams of NOT, NAND, and NOR gates respectively

The design of more complex gates follows on logically from this (Figure 1). To design the NOR gate a second JFET can be connected to the voltage follower in parallel with the first JFET which provides two inputs both with equal function. This means that if either JFET has a voltage applied, the output of the logic gate is the same. The NAND gate was developed using similar reasoning. If both inputs are required to be on to turn the output off, connecting two JFETs in series to the voltage follower will require that both JFETs are sinking current for the output to change. Condensed boolean gates are gates that have an entire Boolean expression compressed into a single gate in order to optimise the number of transistors used (Figure 2). For example, a NOR gate in which one input JFET is replaced by two JFETs in series to ground produces $\overline{AB + C}$ while saving two devices compared to the equivalent NAND/NOR composition. This is analogous to the use of complex gates in integrated-circuit logic design, where several Boolean operations are collapsed into a single transistor network to reduce device count.

The nominal electrical parameters used throughout this paper are summarised in Table II. NAND and NOR gates were built in the lab and compared to the simulated devices, characterised from 1 Hz up to a maximum of 1 MHz; results are presented in Section IV.

TABLE II: Nominal SiC JFET RTL simulation parameters, as defined in `cpus/4004/config.py`.

Parameter	Value
Positive supply V_{POS}	$+24\,\mathrm{V}$
Negative supply V_{NEG}	$-20\,\mathrm{V}$
Logic high	$-0.8\,\mathrm{V}$
Logic low	$-3.6\,\mathrm{V}$
Threshold (digital analyser)	$-2.5\,\mathrm{V}$
Pull-up resistor R_1 (INV/NAND2/NOR2)	$50\,\mathrm{k\Omega}$
Voltage-follower upper R_2	$1\,\mathrm{k\Omega}$
Voltage-follower lower R_3	$4.5\,\mathrm{k\Omega}$
JFET gate–source / gate–drain C_{gs0}, C_{gd0}	$16.9\,\mathrm{pF}$
Simulation temperature	$27\,^{\circ}\mathrm{C}$
Target clock frequency	$100\,\mathrm{kHz}$

B. CPU Design

The Intel 4004 decomposes into six subsystems, each captured in LTspice and verified in isolation before integration: a 4-bit *ALU* (Figure 3) consisting of a 4-bit ripple-carry adder, accumulator and temporary registers, a separate incrementer for scratch-pad register updates, and a two-stage carry flag that breaks the write-back / read race between sum and carry; an *instruction register and decoder* that demultiplexes the high 8 bits of each fetched word into a one-hot instruction line and AND-combines it with a micro-instruction step counter to drive subsystem control lines across three named phases ($A_{1,2,3}$, $M_{1,2}$, $X_{1,2,3}$); a *program counter* with a dedicated 12-bit datapath to a 3-level subroutine *stack*, since the four-bit shared bus cannot move 12 PC bits inside the three-clock execute window; a *sixteen-register 4-bit scratch pad* with pair-mode and individual-word addressing modes; a 4-bit external *bus* treated as a multi-input OR rather than tri-state because the JFET RTL voltage levels do not implement high-impedance cleanly; and a set of *external pins* for data, four CM-RAM signals, CM-ROM, and SYNCH. Per-subsystem JFET counts are given in Table III. Full transistor-level schematics for every subsystem, the carry-stage race-condition derivation, and the micro-instruction phase enumeration are given in the supplementary material.

III. Method

Validation proceeded hierarchically: primitive gates were first checked in LTspice, then composed into sequential sub-circuits and CPU subsystems before full-processor integration. The first sequential subcircuit was the 1-bit synchronous D-type flip-flop. The synchronous part of this simply means that it only latches the D value in on the clock pulse. First the flip-flop was built using discrete logic gates, however, it was noticed that some gates could be reduced using condensed boolean gates as described in the logic primitives section (Section II-A). This was done until the circuit could not be condensed any further due to race conditions and the path for individual signals becoming unbalanced compared to other paths through the circuit. This unbalancing occurred due to the layout of the logic circuit for a D-type flip-flop. The output is taken from a pair of cross-connected gates meaning that if one gate was reached by the signal significantly before the other gate, the output would not latch correctly (Figure 4).

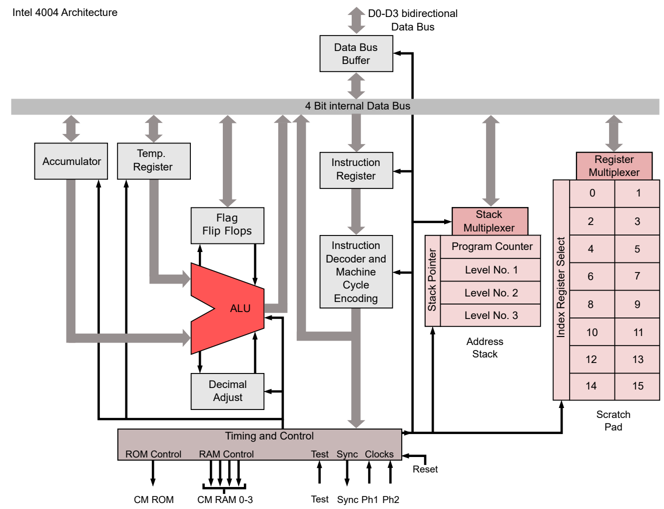

Fig. 3: Architecture of the Intel 4004 [16] showing the layout of registers and connections between CPU sub-systems. Full transistor-level schematics for each subsystem are in the supplementary material.

D flip-flops were used for storage in this CPU as 6-transistor (6T) static RAM (SRAM) and 1-transistor (1T) dynamic RAM (DRAM) proved challenging to implement. There are 146 bits of storage in the 4004; with the current 15-JFET D flip-flop, this accounts for 2190 of the processor's transistors. The full per-subsystem JFET budget of the implementation is given in Table III; the scratch pad and stack together consume 1563 and 817 JFETs respectively, dominating the total of 5537. This total is regenerated from the released schematic metadata using `tools/count_transistors.py`, which walks the `SYMBOL njf` declarations in every component file referenced by the CPU configuration. The count differs slightly from the preliminary hand count of 5328 in earlier reports of this work, which understated the scratch-pad and stack subsystems. Development of 1T DRAM would, by inspection, reduce the total transistor count by roughly 2000 (excluding any RAM refresh circuitry), making storage technology the highest-leverage target for further reduction.

Each sub-system of the CPU was built separately at first in order to decrease simulation times. The system was tested using the following method. Each control line was connected to a voltage source utilising a Piecewise linear (PWL) file. A PWL file is a type of file used by LTspice which specifies the voltage at specific time points. Each control line was listed in a spreadsheet along with register values if appropriate. The necessary voltage level for each control line was then plotted on a timeline to "program" the circuit to perform operations. The PWL files were then generated using Matlab from the list of times the control line needed to be high. The final state of the sub-system could then be compared to the expected state calculated from the spreadsheet. Once each sub-system had been verified as correctly functioning they were assembled into the full CPU by means of a Python script which collated each sub-system and renamed any duplicated component names. This was then tested by connecting the

TABLE III: Transistor count of the SiC JFET 4004-compatible processor, per subsystem at full capacity (all sixteen scratch-pad registers and three stack levels included). Numbers are derived directly from the LTspice schematics referenced by `cpus/4004/config.py`; re-running `tools/count_transistors.py` regenerates this table from the current schematic state.

Subsystem	JFETs	% of total	Resistors
Arithmetic logic unit	566	10.2%	504
Instruction register	680	12.3%	642
Micro-instruction decoder	877	15.8%	675
Control logic	28	0.5%	24
External pins and bus	140	2.5%	159
Program counter	503	9.1%	429
Step counter	363	6.6%	351
Scratch pad (16 $\times$ 4-bit)	1563	28.2%	1230
Subroutine stack (3-level)	817	14.8%	693
Total	**5537**	**100.0%**	**4707**

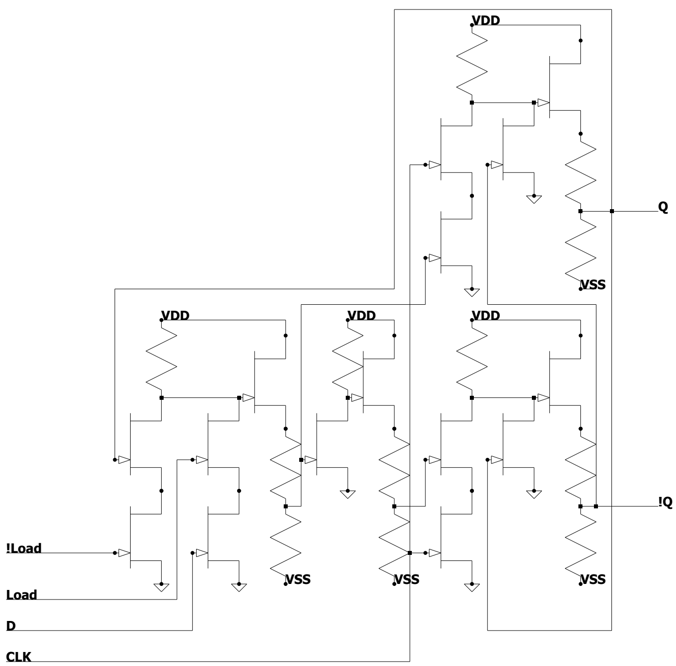

Fig. 4: Circuit diagram for synchronous D-type flip-flop with transistor reductions using combined boolean logic gates

bus to voltage sources and emulating the ROM by feeding it data when it expected to receive data from the ROM. This approach was sufficient for deterministic subsystem tests, but could not validate programs whose control flow depended on CPU-computed state, since every bit returned to the CPU had to be hard-coded beforehand.

A. Assembler

An assembler converts the mnemonics of the instructions from text to binary and converts the operands from hexadecimal or decimal to binary. Thus `FIM 1 0 11` is converted to `0010 0010 0000 1011`. The assembler is a two-pass implementation. The first pass walks the source text and records, for each label, the ROM byte address at which the next instruction will be emitted, advancing a running address counter by one byte per single-word instruction and by two bytes per two-word instruction (`JCN`, `FIM`, `JUN`, `JMS`, `ISZ`). The second pass emits the byte stream, with each label reference resolved to the recorded 12-bit ROM byte address rather than to a source-text line number. The assembled program is written to a `.bin` file – one 8-bit ROM word per line, in ROM-address order – and to a parallel `.hex` file. The assembler is written in Python.

B. PWL File Generation

The `.bin` file is used by another Python script to generate the PWL files that populate the voltage sources inside the 4001-style ROM model. Each PWL file encodes the fixed contents of a single ROM bit cell – either a constant logical high (-0.8 V) or a constant logical low (-3.6 V) – and is named `0.txt`, `1.txt`, `...` in ROM-bit order. Address decoding inside the ROM model selects the addressed byte during the simulated M_1/M_2 phases, so branch and subroutine behaviour during simulation is determined by the CPU state and the ROM address path rather than by an externally scripted bus trace; the PWLs do not encode a precomputed execution trace.

C. Testing and Benchmarking

With a full 4001 and a full 4004 modelled in LTspice, complete programs can be run and verified directly against the documented 4004 instruction-set semantics. The verifier (`tools/verify_micro5.py`) samples the CPU state vector $(\mathrm{ACC}, \mathrm{CY}, R_0, \ldots, R_{15})$ at the end of every `Micro5` phase, where the instruction register has fully settled. Each pair of consecutive samples constitutes a (*pre*, IR, *post*) triple. For every triple the verifier applies the ISA rule for the observed IR — for example, $\mathtt{ADD\ R_n}$ requires $\mathrm{post.ACC} = (\mathrm{pre.ACC} + \mathrm{pre.}R_n + \mathrm{pre.CY}) \bmod 16$ and $\mathrm{post.CY} = \mathbb{1}[\mathrm{pre.ACC} + \mathrm{pre.}R_n + \mathrm{pre.CY} > 15]$, $\mathtt{XCH\ R_n}$ requires that ACC and R_n swap, and so on for all single-effect opcodes (LDM, LD, INC, CLB, CLC, STC, CMC, CMA, IAC, DAC, TCC, TCS, RAL, RAR). A triple that satisfies the ISA rule is a *pass*; one that does not is a *fail* and is logged in the per-instruction diff. Failures whose only inconsistency is the pre-instruction CY bit being flipped (a documented one-sample sampling glitch at instruction boundaries) are categorised separately as *CY artefacts* rather than counted as ISA disagreements.

This methodology has no Python reference emulator running in lockstep, no program-counter tracking, and no sync-point or resync semantics. Every check is local to a single instruction boundary and uses only LTspice-observed pre and post states; a divergence at one instruction therefore cannot cascade into spurious divergences later in the trace. Two test programs are run: the bitwise AND subroutine from the MCS-4 programming manual [12] (a 30-byte routine that exercises XCH, RAL/RAR, ADD, JCN and JMS/BBL — Figure 5) and a 16-bit floating-point multiplication whose execution trace covers 451 dynamically executed instructions and exercises the full ALU through a deeper subroutine call tree. Section IV presents the resulting per-program verification summaries.

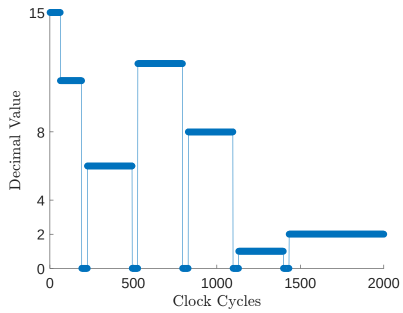

Fig. 5: Figure showing the decimal value in Scratch Pad Register 0 for the operation 11 & 6. This illustrates the first input (11) being shifted left 4 times to compare the highest order bit to the second input (6) with the final result of 2 being loaded in just before 1500 clock cycles

1) Waveform Analyser and Reference Emulator: The LTspice `.raw` file is parsed directly by a Python tool (`tools/trace_synced.py`) which thresholds the per-bit register node voltages at $-2.5\,\mathrm{V}$ to recover digital values, samples them at 95% through each `micro1` phase (after the IR and accumulator have settled), and emits one nibble per register per instruction boundary. A 4004 ISA-level reference emulator written in Python (`tools/rom_emulator.py`) is run in lockstep on the same machine code. Implementation quirks of the original 4004 — the incrementer not affecting the accumulator, the LSB convention for FIN/JIN, the rotate-through-carry semantics — were validated against an online JavaScript emulator [17] during development and against the contemporary MCS-4 manual [12].

D. Experimental Validation of Simulation Models

Physical validation was performed on NAND and NOR primitives implemented using the available SiC JFET integrated circuit (IC) and a copper circuit board with through-hole components. These were connected to a power supply, a function generator, and an oscilloscope. They were then tested at a range of frequencies from 1 Hz to 1 MHz. The data from this experiment was then used to characterise the gates. The inputs from the function generator were taken and a Matlab script was written to convert these into a PWL file which could then be used as the inputs for the simulated gates to get an exact comparison between experimental and theoretical.

IV. Results and Discussion

Table IV collects the headline results of the work. Each row is detailed in the subsections that follow, and the limitations of each claim — in particular the legacy nature of the NAND/NOR measurements and the room-temperature device model — are discussed in Section V.
The primitive-gate physical evidence reported separately in Section IV below covers the NAND and NOR primitives only and supports the gate-level LTspice model rather than the full processor.

TABLE IV: Direct ISA-rule verification of the LTspice CPU trace for each test program. For every (*pre*, IR, *post*) triple observed at consecutive `Micro5` boundaries, the documented 4004 semantics for the observed IR are applied to *pre* and the result compared to *post*. “Instructions checked” is the number of triples; “ISA failures” is the number at which the rule was violated (excluding the separately tracked one-sample CY-input sampling artefacts). No Python reference emulator is run in lockstep, no PC is tracked, and no sync or resync is required, so each verification is local to its own instruction boundary.

Program	Bytes	Dyn. instr.	Instructions checked / ISA failures	Final resu
AND subroutine	30	TBD	TBD / TBD	$R_0 = $ 00
FP multiplication	51	454	454 / 0	5.1875

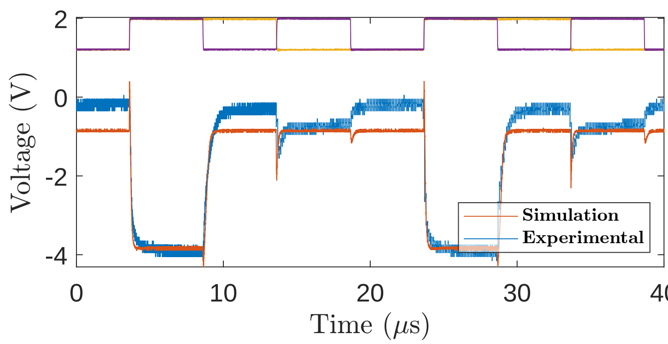

Fig. 6: Experimental and simulation output voltages of a NAND gate at 100 kHz, the operational frequency of the CPU as simulated in this paper. Input voltages are scaled and offset to the top of the figure. Raw oscilloscope traces from the original measurement setup are not retained; the 1.5–2 V baseline shift between measurement and simulation is discussed in Section V.

A. Program Testing

1) AND Subroutine: The AND subroutine from the MCS-4 programming manual was run on the CPU and its register state compared to the Python emulator. Figure 5 shows Scratch Pad register 0 over the course of the routine: the input 1011 (= 11) is loaded, shifted left four times (1011 $\rightarrow$ 0110 $\rightarrow$ 1100 $\rightarrow$ 1000), and the final value 0010 (= 11 & 6 = 2) is written back at $\sim$1400 clock cycles. The initial value of 15 reflects the power-up state of the scratch-pad flip-flops. Full assembly listings, expanded register traces, and supplementary waveform comparisons are provided in the supplementary material.

2) Floating Point Multiplication Subroutine: The floating point system implemented takes the following form.

$$(-1)^{\text{signbit}} \times 1.sssssss \times 2^{eeeeeee-63} \qquad (1)$$

where s denotes a significand bit and e denotes an exponent bit. The processor-level functionality is further demonstrated by the computation 1.4375×3.609375, which gives 5.1875 after rounding in the implemented floating-point representation.

The implemented floating-point format rounds the exact product $5.18847\ldots$ to 5.1875, as shown in Table V. A full description of the algorithm is given in the supplementary material.

B. Experimental Data

The logic gates were tested at input frequencies from 1 Hz up to 800 kHz (NOR) and 1 MHz (NAND). Figure 6 shows the 100 kHz operating point at which the CPU is simulated, and Figure 7 shows the highest frequency at which the NOR primitive still produced an adequate output swing. Per-frequency NAND and NOR comparisons across the full 1 Hz–1 MHz range, including current-draw plots, are given in the supplementary material. Above 800 kHz the NOR output swing collapses because the rise time exceeds the half-period, and the gate can no longer respond to its inputs. For the tested two-input primitives, the NOR gate is therefore the measured limiting case, with adequate output swing up to 800 kHz; the NAND gate continues to switch cleanly to 1 MHz. Both bounds are comfortably above the 100 kHz simulated CPU clock.

TABLE V: Floating Point Number Values Used in Subroutine

Binary	Decimal	Exponential Form	Value
10111000 00111111	184 63	1.4375×2^0	1.4375
11100111 01000000	231 64	1.8046875×2^1	3.609375
10100110 01000001	166 65	1.296875×2^2	5.1875

The primitive-gate measurements show that the measured SiC JFET RTL gates switch comfortably above the 100 kHz CPU simulation clock; this should not be read as a measured full-processor $f_{\max}$, since the complete CPU was not fabricated and the critical paths are only evaluated in LTspice. For reference, the original Intel 4004 silicon was specified at $\sim$740 kHz peak, with the commonly quoted 400 kHz referring to the typical operating frequency rather than the maximum.

The oscilloscope probes introduced scaling by a factor of two, necessitating post-experimental adjustments. There was a 1.5 - 2 V baseline shift in the data compared to what was expected from the simulations, the origins of which are currently under investigation. Some of the difference was due to resistor inaccuracies, however, this cannot account for the entire 2 V shift. Further, it can be seen that as the frequency increases this offset increases, starting at approximately 1.5 V at 2 Hz going up to approximately 2 V at 100 kHz. This indicates that it is some function of the JFETs themselves. This change in output voltage levels as a function of frequency was only observed in the NAND gate. The NOR gate retained its 1.5 V shift for the entire range of examined frequencies. This could be due to differences in the JFETs used between the NAND and NOR gates rather than due to gate topology. Further investigation into the nature of these discrepancies is necessary to determine their source, be it manufacturing variability or the LTspice model itself. The voltage levels and resistor values used should be explored in greater depth to determine optimum performance for specific applications.

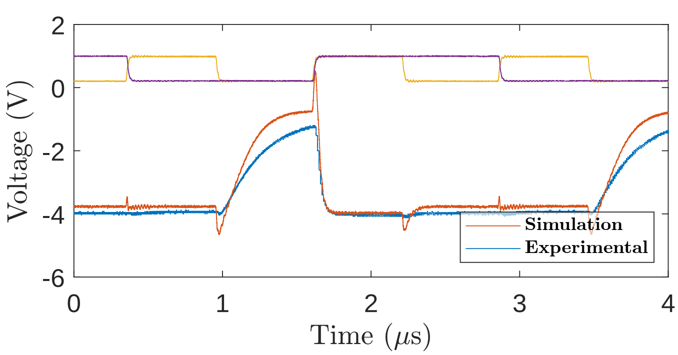

Fig. 7: Experimental and simulation output voltages of a NOR gate at 800 kHz, the maximum frequency at which the NOR gate produced adequate output peaks. Inputs scaled and offset as before. Raw oscilloscope traces are not retained; the 1.5–2 V baseline shift is discussed in Section V.

V. Limitations

Three classes of limitation deserve explicit acknowledgement.

Measured primitive gates exhibit a 1.5–2 V baseline offset relative to the simulated waveforms (Figures 6, 7). The offset is approximately constant for the NOR gate across the entire measured frequency range, and grows mildly with frequency for the NAND. Resistor tolerance is too small to account for the magnitude; this points instead either to manufacturing variability of the SiC JFET die used (a single integrated circuit was the source of devices for both gates) or to a residual modelling gap in the JFET parameter card used for the LTspice simulation. The waveform *shape* (transitions, settling, and frequency response) matches the simulation closely over 1 Hz to 800 kHz for the NOR and 1 Hz to 1 MHz for the NAND, which is the basis on which the simulated CPU is asserted to be functionally representative; the absolute offset is an open discrepancy and should not be read as a confirmation of the simulation's quantitative voltage levels.

The full processor is not fabricated. All system-level claims rest on LTspice transient simulation of the complete 5537-transistor design, cross-validated against an independent Python reference emulator at every instruction boundary (Section III). The strongest experimental evidence in this paper is for the primitive gates only. A fabricated CPU is the obvious next step, but the dominant transistor budget — D flip-flop storage rather than combinational logic — would benefit substantially from a single-transistor DRAM cell before fabrication is attempted.

The LTspice device model used is at room temperature. The CPU simulations use a SiC JFET parameterisation supplied for this project, evaluated at 27 °C; high-temperature behaviour is by extrapolation only. The parameter card itself appeared in the publicly released MEng report from which this work descends and is shipped with the reproducibility package on that basis; the underlying device model belongs to its original owner, and any redistribution beyond academic reproduction should obtain permission accordingly. The radiation-hardness and high-temperature motivation for SiC JFET computing remains; what this work demonstrates is that an Intel-4004-compatible architecture is feasible in a SiC JFET RTL logic family, not that the specific design operates at the temperatures motivating it. A temperature-scaled re-simulation of the worst-case combinational chain and the noise-margin corner sweep is a natural follow-up, deferring the chip-level fabrication question while strengthening the temperature claim.

A full fanout / noise-margin / critical-path characterisation of the integrated design remains future work. The 100 kHz CPU claim rests on LTspice transient execution of the reported test programs rather than on a systematic worst-case critical-path extraction across all 46 instructions, and the primitive-gate frequency results in Section IV should not be read as a substitute. The corner-sweep test benches that would produce noise-margin and timing tables – under $\pm 20\%$ resistor variation and ± 2 V supply variation – are included in the reproducibility package (`analysis/sweeps/`) but not yet executed; the corresponding result tables will be added in revision.

VI. Conclusion

This paper has presented a transistor-level SiC JFET implementation of an Intel-4004-compatible 4-bit microprocessor, comprising 5537 JFETs (Table III) across the ALU, instruction register and decoder, program counter, three-level stack, and sixteen-register scratch pad; a compatible 4001-style ROM model is used for program execution and is not included in this JFET count. The simulated CPU design is supported by two independent forms of evidence: register-level program co-validation against a Python reference emulator at every instruction boundary, demonstrated on a 30-byte bitwise AND subroutine and a 16-bit floating-point multiplication producing a 451-instruction execution trace; and measured waveform-shape and switching-frequency consistency of NAND and NOR primitives built from a discrete SiC JFET integrated circuit, characterised from 1 Hz to 800 kHz (NOR) and 1 MHz (NAND).

The largest practical constraint on further work is the storage cost of D flip-flops: 146 bits of scratch-pad and stack storage account for 2190 JFETs of the 5537 total. The design therefore remains substantially larger than the original NMOS 4004, primarily because the present storage implementation uses 15-JFET D flip-flops rather than 1T DRAM cells. A SiC JFET DRAM cell would close most of this gap. The remaining open discrepancy in this paper — the 1.5–2 V baseline offset between measured and simulated primitive gates — is a candidate next investigation, and is most cheaply pursued by repeating the NAND/NOR measurement with multiple device samples to separate manufacturing variability from a residual modelling gap. A temperature-scaled re-simulation of the worst-case combinational chain and the noise-margin corner sweep is a complementary follow-up that strengthens the high-temperature claim without requiring fabrication. The fabrication of the full 4004-compatible processor in SiC JFET is the eventual target; the work presented here establishes that a complete architecture-scale design and a validation methodology now exist to support that step.

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